

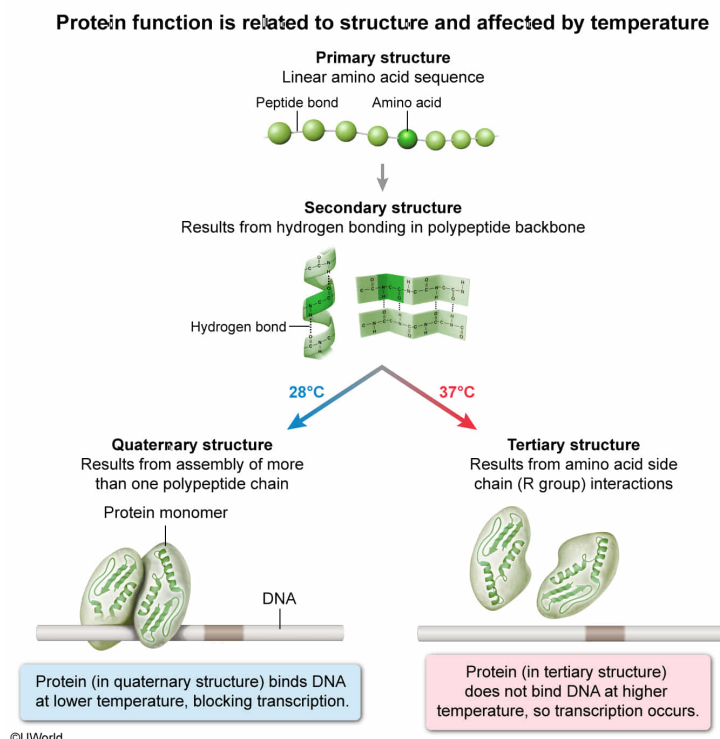
A protein found in *Salmonella* bacteria can exist as either a dimer that binds DNA and represses transcription of the gene encoding the protein or as a monomer that does not bind DNA. At lower temperatures (28 °C), the bacteria expressed the protein at lower levels than at higher temperatures (37 °C). These findings are most likely evidence of:

- ☐ A. disruption of primary but not tertiary structure at 37 °C. [5%]
- ✓ ☒ B. disruption of quaternary but not primary structure at 37 °C. [58%]
- ☐ C. alteration of both secondary structure and quaternary structure at 37 °C. [16%]
- ☐ D. alteration of neither secondary nor tertiary structure of at 37 °C. [19%]

Correct

58% answered correctly

Explanation:



Peptide bonds link amino acids, and thereby form proteins. The sequence in which amino acids are linked defines a protein's primary structure and, along with the chemical characteristics (eg, side-chain charge) of the amino acids in the sequence, determine protein structure and function. Environmental factors can change protein structure (eg, denaturation) and, consequently, protein function.

The question describes a protein in *Salmonella* bacteria that can exist as a dimer functioning as a transcription factor that represses transcription or as a monomer that does not bind DNA. At lower

temperatures, less protein was produced, indicating that transcription was repressed. Thus, lower temperatures most likely permit dimerization and subsequent repression of transcription. At higher temperatures, dimerization does not occur and transcription increases. Therefore, the experimental results provide the strongest support for the statement that **quaternary structure is influenced by temperature** and is **disrupted at 37 °C**.

(Choices A, C, and D) The given information suggests that quaternary structure of the protein is disrupted at 37 °C, but no conclusions can be drawn about the effect of temperature on tertiary, secondary, or primary structure.

Educational objective:

A protein's environment and amino acid sequence determine its structure and function.

Time spent:0 sec

Subject:
Biochemistry

QID:403537

System:
Amino Acids and Proteins